ARCHITECTURE & PLATFORM OVERVIEW

DeliveryRouter

High-Performance Enterprise Fleet & Logistics Optimization Platform

Microservice-driven route optimization, real-time spatial tracking, and granular fleet payload management.

TECH STACK ECOSYSTEM

Laravel 12 | Python FastAPI | Docker Compose | Livewire 3 | Redis | Leaflet.js

Khder Jaber Ahmed
Backend & Software Engineer • PHP/Laravel, Python Microservices, System Architecture

Production-Ready Build

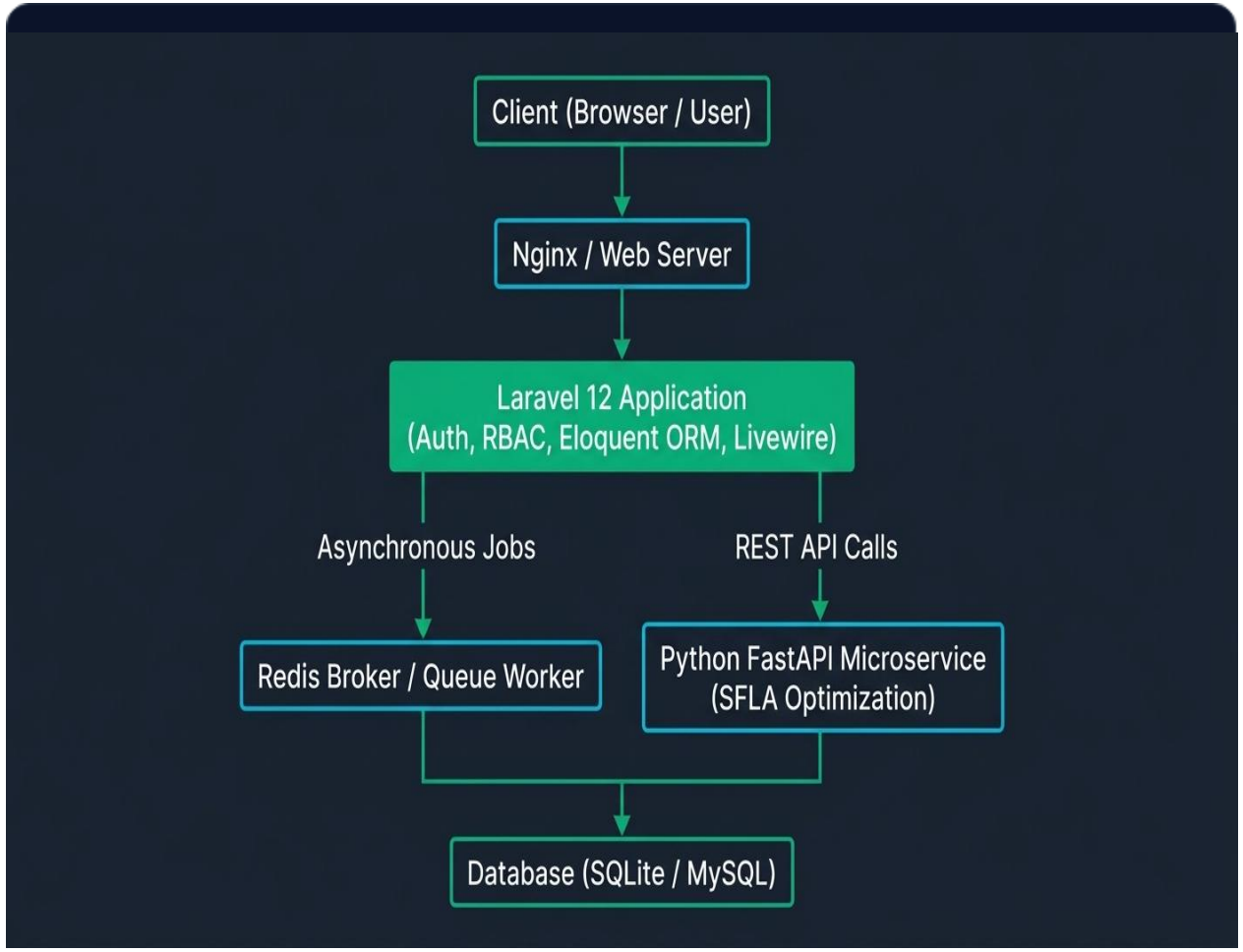
ENGINEERING MATRIX

Legacy Bottlenecks vs. DeliveryRouter Solutions

Legacy Monolithic Bottlenecks	DeliveryRouter Architectural Solution
High Latency: Synchronous O(N^2) VRP route calculations freeze HTTP workers.	**Asynchronous Decoupling:** Heavy routing algorithms offloaded to dedicated Python engine.
Monolithic Overload: Single-threaded PHP processes struggle under spatial computation.	**FastAPI Microservice:** Asynchronous ASGI service handling distance matrix execution.
Unoptimized Payloads: Lack of dynamic payload validation leads to overloaded fleets.	**Capacity Constraints:** Real-time max payload math preventing fleet overloads.
Page Reloads: Slow GIS map re-renders disrupt dispatch operations.	**Reactive GIS:** Livewire 3 + Alpine.js with Leaflet.js canvas updates without reloads.

SYSTEM DESIGN

Microservice Architecture & Data Flow



Laravel 12 Domain Core

Handles RBAC, business logic, session security, and state mutations.

FastAPI Optimization Engine

Executes Shuffled Frog Leaping Algorithm (SFLA) for dynamic VRP solving.

Redis Event Queue

Manages asynchronous jobs, preserving sub-50ms HTTP response cycles.

Isolated Docker Stack

Orchestrates laravel_app, fastapi_app, and redis on a private bridge network.

FUNCTIONAL SUITE

Production Core Modules

Analytics Dashboard

Real-time fleet tracking & KPI monitoring via Chart.js with non-blocking background polling.

Order Lifecycle

Strict finite state machine (Pending → Dispatched → Delivered) with spatial coordinate indexing.

GIS Route Planner

Leaflet.js & OpenStreetMap mapping active depots, vehicles, and distance matrices.

Capacity Controller

Dynamic vehicle state switching and strict payload weight bounds enforcement.

Dispatch Hub Manager

Central hub node configuration with spatial order grouping based on proximity.

Granular RBAC

Role-Based Access Control enforcing Admin, Manager, Driver, and Customer scopes.

RELIABILITY & HARDENING

Performance, Security & DevOps Engineering



Defensive Security

Rate-limiting middleware on auth routes mitigating brute-force attacks. Enforced encrypted HTTP-Only session cookies with strict CSRF protection.



Query Optimization

Eager loading on Eloquent relationships eliminating N+1 query overhead. Optimized spatial indexing for high-frequency geolocation lookups.



Fail-Safe Containerization

Clean separation of concerns with isolated Docker containers. Circuit breaker patterns for seamless API communication between PHP and Python.

DEPLOYMENT & NEXT STEPS

Get Started with DeliveryRouter

QUICKSTART COMMANDS

```
git clone https://github.com/KhderAhmed/Better-route-tracking-system
 cd Better-route-tracking-system
 docker compose up --build -d
```

Open for Roles

Software Engineering / Senior Backend Roles

Khder Jaber Ahmed

Reach out on LinkedIn for technical walkthroughs